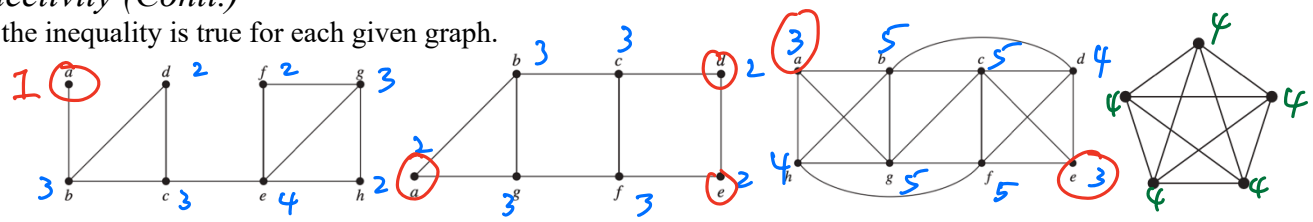
MAT2540, Classwork18, Spring2026

10.4 Connectivity (Conti.)

15. Check if the inequality is true for each given graph.



$\kappa(G)$	1	2	3	4
$\lambda(G)$	1	2	3	4
$\min_{v \in V} \deg(v)$	1	2	3	4

16. *Definition:* The **Path** of a **Directed Graph**. Let n be a nonnegative integer and G a directed graph.

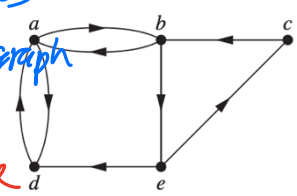
A *path of length* n from u to v in G is a sequence of edges $e_1, \dots, e_n$ of G such that e_1 is associated with (x_0, x_1), e_2 is associated with (x_1, x_2), and so on, with e_n is associated with (x_{n-1}, x_n), where $x_0 = u$ and $x_n = v$.

When there are no multiple edges in the directed graph, this path is denoted by its vertex sequence $x_0, x_1, \dots, x_n$.

A path of length greater than zero that begins and ends at the same vertex is called a *circuit* or *cycle*.

A path or circuit is called *simple* if it does not contain the same edge more than once.

17. Example: Path of a Directed Graph.



(1) a, b, e, c, b is a path of length 4 since (a,b), (b,e), (e,c), (c,b) are edges in this graph

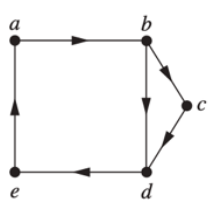
(2) a, d, a, d, a is a path of length 4 since (a,d), (d,a) are the edges, but not simple

(3) a, d, b, e, a is not a path since (d,b) is not an edge
(e,a) is not an edge

18. *Definition:* The **Strongly Connected** Graph and **Weakly Connected** Graph

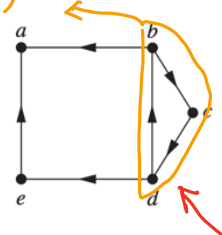
A directed graph is strongly connected if there is a path from a to b and from b to a whenever a,b are vertices. A directed graph is weakly connected if there is a path bewteen a and b in the underlying undirected graph.

19. Example. Check if each given graph is strongly connected or weakly connected.

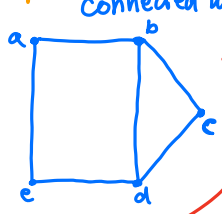


Strongly connected.

strongly connected component



connected when it is an undirected graph



then it is weakly connected.

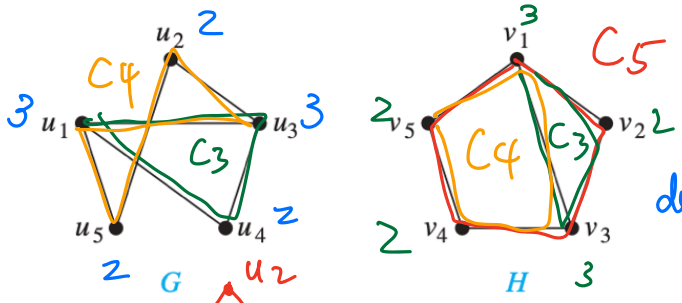
Note: Clearly, any strongly connected graph is weakly connected.

20. *Definition:* The **Strongly Connected Components**.

The subgraphs of a directed graph G that are strongly connected but not contained in larger strongly connected subgraphs, that is, the maximal strongly connected subgraphs, are called the strongly connected components or strong components of G.

21. Paths and Isomorphism.

Besides three invariants (number of vertices, edges, and degree sequence), the *path* provides a further look on two graphs to see if they are isomorphism.

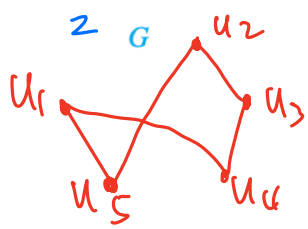


G has 5 vertices H has 5 vertices

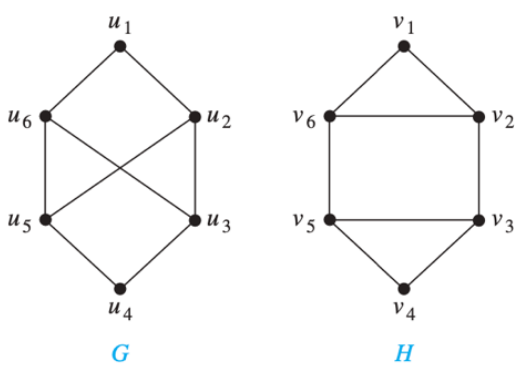
G has 6 edges H has 6 edges

degree sequence of G: 3, 3, 2, 2, 2

degree sequence of H: 3, 3, 2, 2, 2



isomorphism f:

$f(u_1) = v_1$, $f(u_2) = v_4$, $f(u_3) = v_3$

$f(u_4) = v_2$, $f(u_5) = v_5$